

# Purnendu Tiwari

Rewa, India | +91 9685021295 | purnendutiwari2004@gmail.com  
linkedin.com/in/purnendutiwari | github.com/purnendu5804

## Experience

### Software Development Intern

May 2026 – July 2026

FOSSEE, IIT Bombay

Mumbai, India (Remote)

- Collaborated with a mentor and a 3-member team to modernize the **Linux packaging pipeline** for the eSim application across **AppImage** and **Flatpak** formats
- Fixed **PyQt5** dependency crashes by bundling a self-contained **Python runtime** into the AppImage, removing all host-side setup requirements
- Resolved **Qt platform plugin** failures by building automatic **X11/Wayland** detection logic, fixing rendering across Linux desktop environments
- Rebuilt the **Flatpak** offline install pipeline by fixing broken package sources and pre-validating dependency wheels, enabling zero-network builds

## Projects

### sdcHouse – Real-Time Virtual Space Platform | *Next.js, Node.js, Socket.IO, WebRTC*    **Live | GitHub**

- Built a **real-time multiplayer web application** enabling synchronized avatar movement and interaction across concurrent connected clients
- Implemented a **Socket.IO-based signaling server** on **Node.js/Express** to manage global state (player positions, chat) and broker **WebRTC** peer-to-peer connections
- Developed **peer-to-peer audio/video communication** using **WebRTC**, avoiding server-side media relay for scalable private interactions

### RCE Engine – Remote Code Execution Platform | *Docker, PostgreSQL, Redis, GitHub Actions*    **GitHub**

- Architected a **Remote Code Execution engine** inspired by Judge0, executing untrusted code in **resource-limited, network-isolated Docker containers** with automated timeout enforcement and cleanup
- Designed an **asynchronous job pipeline** using **BullMQ** and **Redis** to queue, process, and poll concurrent code submissions with horizontally scalable workers
- Built **RESTful APIs** with **Express** and **Prisma/PostgreSQL** to persist submission state and manage container lifecycles reliably end-to-end
- Engineered a **CI/CD pipeline** with **GitHub Actions**, including type-checking, Docker image validation, and integration tests against live Postgres/Redis service containers

## Technical Skills

**Languages:** Java, JavaScript, TypeScript, Python, SQL, Bash

**Web Development:** React JS, Node JS, Express JS, REST APIs, MongoDB, MySQL, PostgreSQL, Next.js, Prisma

**Cloud & DevOps:** AWS (S3, CloudFront, IAM, VPC, EC2), Docker, Terraform, GitHub Actions, CI/CD, Prometheus, Grafana, Cloudflare, Linux

**Developer Tools:** Git, GitHub, Bitbucket, Jira, Redis, BullMQ, Postman

## Education

### Vellore Institute of Technology (VIT)

Bhopal, India

*B.Tech in Computer Science; CGPA: 8.97/10.0*

*2023 – 2027*

## Certifications

**AWS Certified Solutions Architect – Associate (SAA-C03)** – Amazon Web Services

**Introduction To Machine Learning** – NPTEL (IIT Madras)

## Achievements & Co-curricular

**Top Coder:** Cleared inter-campus qualifier, securing selection among the top 363 out of 10,000+ students

**Leadership:** President, Software Development Club; led a 6-member team to host 5+ tech events for 300+ students

**LeetCode Knight:** Achieved Highest Contest Rating **1852** and Among the Top 5% of coders on LeetCode

**GirlScript Summer of Code:** Ranked **419th** among global open-source contributors

**National Debate:** 2nd Runner Up, Ek Bharat Shresth Bharat Competition

**Music:** Diploma in Classical Music (Raja Mansingh Tomar Music Arts University)